CHAPTER 10

Administration and Operations

THE US HEALTHCARE ECONOMY is valued at over $4 trillion. If it was a country, it would have the fifth largest economy in the world, and it equates to nearly half of what the entire world spends on healthcare. We can argue about whether that's a good thing or not and what percentage of it is actually necessary, but we can all agree that it's an inefficient system. It's convoluted and the overall setup creates redundancies and needs lots of approvals and what we used to call paperwork. In fact, it's estimated that about 25% of this $4 trillion is due to administration, which means that we spend $1 trillion a year on paperwork. It's no wonder that healthcare is now the number one employer, passing retail as of 2017 with 17 million jobs.[1]

Administrative money needs to be spent to support activities outside of direct patient care. Given that most economies in the world are smaller than $1 trillion, you could say that the administrative part of the US healthcare system is also one of the largest economies in the world. Part of that is because we have more than 1000 insurance companies, each with its own policies and forms, requiring a major revenue cycle function in each hospital system, and even in small doctors' medical clinics. A typical medical clinic or health system has to deal with so many different payers that many of its employees are occupied with filling out forms and making calls to ensure that they're meeting each payer's requirements, because that's what pays the bills and keeps the lights on. When we think of care delivery institutions, this isn't what typically comes to mind. Unfortunately, it's a reality because of the disjointed payment system in our country and the lack of regulation to solve the problem.

Another contributing factor is the lack of an efficient source of longitudinal patient medical records which would simplify data sharing and access of information for providers, commonly referred to as a health information

AI Doctor: The Rise of Artificial Intelligence in Healthcare: A Guide for Users, Buyers, Builders, and Investors, First Edition. Ronald M. Razmi.

exchange. A single national healthcare database could be achieved through either a commercial or government program that would ensure that all healthcare providers had access to the same data for patients instead of having to waste resources searching for and retrieving data from other institutions.

All of this means that both payers and providers have to expend tremendous resources on administrative tasks. The good news is that we can expect to see future workflow improvements through robotic process automation (RPA), artificial intelligence (AI), and other promising technologies. These technologies are a natural fit to improve efficiency through automating repetitive tasks with minor or moderate complexity. Most of these administrative functions are done electronically, making AI the perfect tool to "figure out" what to do and to perform activities like completing forms, sending information, and scheduling follow-ups. This would bring three economic improvements to the process: first, it would mean fewer healthcare resources, including the time of doctors and nurses, spent on administration; second, it would enable more timely completion of these steps and reduce the time to achieve revenues; and third, for the same budget, it would enable the allocation of more money for actual patient care. As Rock Health put it in their report about AI in healthcare: "*Media has largely centered on the ability of AI/ML to transform how clinical care is delivered through better diagnostics and treatments. But AI/ML is even more quickly transforming the business of healthcare through the automation and enhancement of non-clinical, operational functions—such as claims adjudication, patient engagement, scheduling optimization, risk analytics and documentation.*"[2]

These AI systems could also help us to reduce burnout among physicians that's brought on by the ever-increasing need for doctors to carry out administrative tasks like coding, claims, pharmacy refills, and preauthorization. Carrying out these manual, repetitive tasks for billing and insurance is frustrating for physicians and costly for healthcare providers. That's why the ability of AI to reduce these tedious and time-consuming workflows in the backend could be one of its biggest impacts on the healthcare industry. It's estimated that AI could lead to time savings of 17% for doctors and over 50% for nurses.

Let's take a closer look at how AI can help.

10.1 Providers

10.1.1 Documentation, Coding, and Billing

The administrative staff of medical institutions deal with a range of revenue-generating activities in support of medical care, like verifying patients' insurance eligibility, identifying the right medical codes based on the services provided, submitting claims to insurers, and following up with patients about outstanding bills. With advances in AI technologies such as computer vision and machine learning-enabled bots, many of these activities can be intelligently automated. RPA, an umbrella term for automating repetitive back-office tasks like onboarding and document digitization, has especially benefited from advances in computer vision and natural language processing. This is very relevant to the major issue of excessive administrative costs. Given that the technology finally seems ready for its moment, more hospitals are weighing the benefits of using AI for coding and billing automation.[3]

Managing billing, revenue collection, insurance filings, and other related activities are at the frontier of AI's entry into healthcare. That's because it's a low-risk area with an immediate financial impact on the health system. It doesn't carry the liability risks of clinical AI applications, where small errors could have large consequences on patient outcomes. Furthermore, medical providers don't need to worry about securing reimbursement, and the business case is much more compelling in the short-to-medium term than clinical AI applications.

More than 40% of submitted claims aren't paid electronically or automatically upon their first submission. The AI applications in this area include more accurate coding from processing provider documentation and identifying the relevant billable medical codes. There are now algorithms that can tap into historical data and use it to predict whether a claim will be approved or denied. The medical insurance industry is increasingly recognizing the potential of AI for improving billing and insurance tasks. There are three main ways that AI is being used to revolutionize healthcare billing and insurance: natural language processing (NLP) for automated, computer-assisted provider documentation (CAPD) and computer-assisted coding; AI for the prioritization of clinical documentation improvement (CDI) and electronic prior authorization; and AI-powered RPA to help with claims management. Generative AI is showing great promise in this area and the large language models (LLMs) at the basis of generative AI will undoubtedly become the basis of AI solutions that will anticipate the words and thoughts of clinicians so that they can further speed access to information and reduce administrative tasks.

Today's AI-powered RPA startups are on a mission to automate the repetitive, manual tasks like claims submission and denial that occur during the end-to-end revenue cycle. In medical coding, players such as Google-backed Nym use NLP to automate the labor-intensive process of translating EHR notes into billable code.[4] Infinitus Systems is developing a "voice RPA" chatbot that asks questions about payment authorization and other procedures and records answers in the relevant fields.

LLMs from Google (Bard) and OpenAI (ChatGPT) have already made impressive progress when it comes to interpreting medical documentation. Healthcare-based NLP tools have also been released on open-source licenses by the National Library of Medicine (part of the National Institutes of Health), Amazon Comprehend Medical, and Google via its Healthcare Natural Language API. Healthcare AI tools for administrative automation will mostly depend on EHR data. There are tools like NYM that tap into EHR notes to translate health services into billable codes, while RPA platforms are using AI to extract data from EHRs to populate insurance claims forms. This is also known as "charge capture," the process by which doctors translate patient visits and diagnoses into medical codes that can be billed to an insurer. Historically, this has been done by hiring people who read the doctor's notes and update the coding for that visit.

Companies such as Apixio and 3M have developed AI-powered risk adjustment and hierarchical condition category auditors to help practices optimize their coding for quality payment programs.[5] We already know that AI does a better job of spotting patterns than human beings. Traditionally, rules-based engines needed to be constantly updated to reflect changes and to stay accurate over time, but AIs can automatically become better at what they do, especially when it comes to pattern recognition.

One of the companies working with CAPD technology is HITEKS, which provides embedded EHR workflows within Epic to provide timely advice on documentation to support ICD-10 claims coding. Their CEO, Dr. Petratos, is a medical informatics-trained physician who talked to me about some of the issues that medical centers are facing and how their NLP technology is solving some of these issues. According to Dr. Petratos, the sophisticated hospital systems in the United States are now seeking to notify their providers of CDI opportunities in near real-time before they sign their notes, and within the workflows of the EHR. Legacy processes, which include a nurse, physician, or coder manually reviewing each chart after provider notes are written (reactive CDI) and sending a note to the provider's inbox, are inefficient.

"Reactive" CDI takes three times as long and can leave an organization vulnerable to denials due to changed documentation flags by insurance. CAPD software that includes advances in accuracy and timeliness can reduce what's commonly known as "query fatigue," when providers see too many alerts in the EHR and start to tune out. HITEKS developed an approach where they combined certain rules, which are known to be clinically relevant and are required for compliance reasons, with machine learning to suppress inappropriate alerts. "Proactive" CDI is the result of these AI advancements and has been measured to improve the bottom line with a 3% increase in revenues from inpatient accounts, 17% reduction in provider administrative time, and 25% reduction in CDI resources to oversee the CAPD processes.

AI's ability to process and understand natural language (e.g. ambient AI) is also being harnessed to help clinicians create documentation by listening to their interactions with patients to create clinical notes reflective of patients' complaints and providers' assessments and plans. Done well, this could allow us to ease the timeliness and provider burden to simplify the creation of more complete and comprehensive documentation, showing the thought processes of providers much more clearly than our existing approach. It can also help to show the decision-making process that the provider followed, helping with coding and minimizing the risk of denials. As we discussed in Chapter 9, AI could also help to reduce the administrative burden that comes from documentation, allowing us to better highlight the full complexity of each patient's case, leading to better coding and fewer rejected claims. Currently, ambient AI has had limited penetration into US healthcare, but LLMs are showing great promise in this area.

One of the key areas on the administrative side of healthcare is prior authorization. This is when clinicians need to get an approval from the insurance company to perform a service like an MRI for the patient. Insurance companies use prior authorization to control costs by preventing unnecessary procedures. This is always a point of contention since doctors think they should be the ones to make decisions about what's appropriate for their patients. However, insurance companies argue that testing and procedures are often ordered by doctors when medical guidelines don't support them. Most recently, doctors have used ChatGPT to generate these letters to the insurance companies. Although a great time-saver, ChatGPT has shown an ability to provide false scientific references in these letters or fabricate patient data. This underlines the fact that although we're heading for exciting days, the road to get there is not fully paved yet. However, one of the early use cases for generative AI is that of generating prior authorization letters that summarize key information from the EHR and explain why procedures are necessary.

Let's use the example of someone who's presenting with back pain for the first time. The overwhelming majority of patients with back pain improve spontaneously without the use of expensive medical procedures like MRIs. However, some clinicians go straight to ordering an MRI as their preferred diagnostic tool. I can see both sides of the argument, but being a physician, I tend to agree that clinicians know best. There are often circumstances that mean the clinician will deviate from the guidelines for good reason. But they need to do a better job of documentation so that insurance companies agree to pay for it. AI can help with the documentation, ensuring that these extenuating circumstances are included on the forms that will be filed with the insurance company. It would also take care of a lot of annoying and time-consuming tasks for the medical staff.

AI can be used to harvest the appropriate information for prior authorizations and RPA can populate the insurance forms with that information to allow for automatic submission with a higher likelihood of approval. This is good news for providers and their employees because it means they don't have to manually complete forms or set up rules for prior authorization. For example, companies like Epic, HealthFortis, and UIPATH use RPA and workflow software to automate important administrative processes such as eligibility checks, insurance claims, prior authorizations, appointment reminders, billing, data reporting, and analytics.[6]

10.1.2 Practice Management and Operations

AI can help providers and their practices improve operations like scheduling and patient check-ins and check-outs. AI-powered scheduling software could make life easier for patients and their medical providers. It will be able to analyze previous scheduling patterns for each patient and provide suggestions for the best date, time, provider, and location for the patient. It can even automatically schedule appointments to further streamline the process.

If a patient needs to receive multiple treatments or to see several providers on the same day, these tools could develop an optimized schedule for them to follow. This reduces the amount of time spent by both the patient and the scheduling staff to create the next appointment or series of appointments.[7] AI tools can also better understand the complexities that come with every patient and every treatment plan, and these scheduling tools can figure out the optimum visit length for each appointment. A visit for a patient with multiple diagnoses should take longer than a visit for a patient who's normally healthy.

This approach allows us to optimize providers' schedules by customizing the amount of time that each patient receives based on their individual needs. Healthcare providers can provide only the amount of time that's necessary without falling behind in their schedule or rushing from one patient to another. Figuring out these estimates and implementing them into a schedule is prohibitively time-consuming for human beings, but AI can handle it in just a couple of seconds. As we discussed, that can include automatic documentation, better coding, completing forms, and more.

This also ties into the concept of risk-adjusted paneling and resourcing. Risk-adjusted paneling can ensure that primary care physicians have adequate time to address the needs of each patient by increasing or decreasing panel sizes based on patient complexity, which can boost patient satisfaction and lead to better work–life balance for physicians.[8] At UCSF, EHR data on healthcare usage has been used to train algorithms to weigh panel sizes in primary care.[9] The hope is that in the future, we'll be able to use models like these to better determine staffing levels for medical practices based on the amount of care that's required.

AI can also be used to automate certain aspects of pre-visit planning to make physician-care encounters more efficient and rewarding for patients and physicians.[6] This would improve the overall workflow at the physician's office as patients can be checked in faster, physicians can view prioritized information to make visits more efficient and follow-ups can be scheduled automatically.

Another way that AI can improve practice management is through the use of chatbots that are more intelligent due to the fact that instead of being programmed, they're trained with AI and machine learning. This helps to ensure that patients' expectations are met and so they'll start using those technologies more often. Chatbots can save the medical staff from having to spend a lot of time answering the same questions over the phone, and this has been used recently for the management of COVID patient calls. Web-based bots were used to answer questions, address vaccine issues, keep people out of ERs and hospitals, and manage them remotely if they were healthy enough.

Patient-facing chatbots trained on medical literature with LLMs and patient data can build a holistic view of a patient's condition using multiple modalities, ranging from unstructured descriptions of symptoms to continuous glucose monitor readings and patient-provided medication logs. After interpreting this heterogeneous data, the models could interact with the patient, providing detailed advice and explanations. Importantly, these models can facilitate accessible communication, providing clear, readable, or audible information on the patient's schedule.

AI-based chatbots perform a number of functions that could improve operations at medical practices. These solutions can centralize resources, offering location-based information on the programs that are available and providing dynamic answers to specific questions about insurance. An obvious advantage of these chatbots is that they're available 24/7. We just need to make sure that they can integrate with the scheduling tools and EHRs that are currently in use.

Botco.ai's chatbot is deployed on medical practice websites to carry out conversations with visitors, decreasing the amount of time that human employees need to spend on the phone. Implemented at Aspire Indiana Health, Botco's tool operates as an extension of many departments, answering questions about topics like insurance coverage, appointment scheduling, telehealth, behavioral health, addiction, employment and housing, recovery services, and residential and outpatient treatment services.[10] It also integrates with their practice management software and EHR and can schedule and modify appointments to save the staff from having to perform those tasks.

Another important area for medical practices and health systems is ensuring compliance with accepted guidelines and requirements from the government and insurance companies. This is especially critical in coding and billing. There's a long history of providers and institutions engaging in the practice of upcoding, using codes that aren't justified by the level of service provided to achieve a higher level of reimbursement. This is an area where criminal investigations and/or civil lawsuits can create headaches for organizations. As such, detecting and addressing it before the government or insurance companies discover it is important for medical organizations.

AI can help with this, both for payers and providers. As well as spotting fraud faster than human beings, it can stop it from happening in the first place. It does this by processing huge amounts of data and identifying upcoding patterns, looking for appropriate documentation, and flagging anything that seems out of place. As with other areas of healthcare, it's not designed as a replacement for existing fraud detection efforts, but rather as an enhancement that can alert human operators whenever they spot something that's cause for concern.

New AI and machine learning techniques can identify code words that are used in fraudulent situations or patterns of medical coding that could indicate fraud. They do this by understanding both the contents of a document and the context around it, which includes the people involved and the relationships between them. It can spot words that aren't normally used or that are repeatedly used only among a small group of participants. By using AI to automate the process, enterprises can save both time and money, while understanding their data and avoiding issues like potential fraud.[11]

10.1.3 Hospital Operations

AI can be instrumental in helping hospitals improve operational decision-making by using quantifiable metrics to assess the quality of healthcare. As with medical practices, RPA plays a key role in streamlining hospital operations. We mentioned earlier companies that are using AI to improve billing and coding for medical practices. Their use of RPA, computer vision, and machine learning to automate healthcare administrative workflows like checking claim statuses and managing accounts receivable adds significant value to hospital operations.[12]

AI optimization can be critical for managing healthcare assets like hospital rooms, operating rooms (ORs), and radiology scans, as well as vaccine administration, bed allocation, and sequencing activities (room cleaning, discharges, etc.). Companies like LeanTaas are using AI to help hospitals to create capacity. To make this a reality, we need electronic data sources and optimal access to different hospital databases, as well as the ability to aggregate and analyze. This may mean redesigning data systems and processes and introducing technology for optimization and forecasting.[13]

There are a million hospital beds in the United States. Room turnover in hospitals is far less predictable than it is in hotels, so it's harder to manage capacity and operations. Prior to being discharged from a hospital, a patient might need to wait for labs, post-surgery observations, and even a discharging physician who's dealing with an emergency. Using sophisticated AI models that are far better than traditional forecasting tools, you can better forecast and redesign processes to unlock significant capacity.[14]

Given that AI solutions can make sense of data in an ongoing manner and feed those insights to a medical facility, they can improve both clinical and operational decisions, which are very much intertwined. If sepsis is a big issue in a hospital, patients stay longer and that affects the availability of beds, along with other hospital economics. If hospital capacity is better managed using AI's predictive capabilities, more beds will be available, which means patients will get admitted and receive treatment much faster, leading to better outcomes. As such, many of the AI solutions that improve hospital operations are also addressing ongoing clinical issues. Examples of those include:

- Optimizing scheduling and staffing tasks
- Improving clinical workflows by scanning documents and providing recommendations
- Automating tasks like reading scans or interpreting results to expedite decision-making
- Using algorithms to reduce scanning time and radiation to the patients
- Using algorithms to expedite image reconstruction
- Using segmentation to improve radiation therapy
- Predicting hospital-acquired infections such as sepsis and *C. diff*
- Using infrared, computer vision, and sensors to reduce nosocomial infections and track handwashing
- Tapping into reinforcement learning to wean patients off ventilators
- Providing decision support for the use of IVF, pressors, and ventilations
- Powering ICU surveillance of patient vitals and other metrics, as well as ICU discharge
- Monitoring conditions in the OR
- Preventing patient falls
- Using remote monitoring and algorithms to pick up on brewing issues such as sepsis or heart failure

Much of this revolves around better outcome prediction, which in turn leads to better resource allocation. The information required for this type of predictive modeling comes from EHR data and hospital bed data, using metrics like hospitalization length, the number of procedures, and the number of patients with sepsis and other nosocomial infections.

There are various examples of algorithms predicting mortality using structured data from EHRs. One such algorithm by Google had a high success rate when predicting mortality, hospital stay duration, readmission rates, final diagnoses, bleeding complications, kidney failure, and more. Algorithms are showing promise in silico for predicting diseases such as hypertension, kidney failure, diabetes, and arrhythmia by using EHR data, including lab results, trends over time, and more.[15]

Thirty percent of hospital patients who fall experience injury (such as fractures and head trauma), and this boosts the length of their stay by an average of 6.3 days. Each fall that results in an injury costs around $14,000. Qventus analyzes call lights, bed alarms, EMRs, meds, vitals, and ages to assess the risk of falls in hospital. Their deployment at El Camino Hospital resulted in a 29% reduction in falls by identifying when the falls usually happen and the sequence of events that lead to them.[16]

CommonSpirit Health, a combination of Dignity Health and Catholic Health Initiatives, is a nationwide health system of more than 350 hospitals, and it's currently using an AI tool from Lean TaaS that helps to optimize the use of OR suites. The tool, iQueue, uses AI to analyze the use of ORs in an institution, identifies unused OR time, prompts surgeons to release unused OR time, and then allows the hospital to make that time available to other surgeons. Since using the tool, CommonSpirit Health has seen a 21% release fill rate, which means that 21% of the identified available time is being used by other surgeons. This translates to an ROI of 14.5× for the health system.[17]

Another key issue in hospitals (and medical practices) is the security of data and the need to easily transfer it to facilitate better medical care for patients while ensuring privacy and security. Companies are already developing AI-driven privacy tools for healthcare administrators to help them to securely transfer patient data. For example, Baltimore's Protenus has developed healthcare offerings that offer auditing, security, and compliance tools that are powered by AI and which can be used for patient and clinical trial data. San Francisco's Datavant allows users to de-identify patient data so that they can exchange it with other healthcare organizations without compromising patient privacy.[3, 12]

Figure 10.1 shows some of the use cases of AI for hospitals and medical practices. In addition to the ones we discussed, AI can also improve marketing, optimize pricing, and provide competitive intelligence to boost the top line for clinics and hospitals.

AI in Healthcare: Healthcare Management

- **Brand management and marketing:** Create an optimal marketing strategy for the brand based on market perception and target segment
- **Pricing and risk:** Determine the optimal price for treatment and other service according to competition and other market conditions
- **Market research:** Prepare hospital competitive intelligence
- **Operations:** Process automation technologies such as intelligent automation and RPA help hospitals automate routine front office and back office operations such as reporting
- **Customer service chatbots:** Customer service chatbots allow patients to ask questions regarding bill payment, appointments, or medication refills
- **Fraud detection:** Patients may make false claims. Leveraging AI-powered fraud detection tools can help hospital managers to identify fraudsters

FIGURE 10.1 *(source: Original Research)*

10.2 Payers

Like providers, payers can reap enormous benefits from using AI to manage their businesses. That's because insurance companies rely heavily on statistics and numeric analysis. They need to accurately estimate what it will cost to cover the medical care of a population; design benefits based on that; charge the right premiums and monitor as the claims keep coming. If they're not on track, they'll be out of business by the end of the year.

If there's one thing that AI does well, it's to take lots of data, figure out patterns and relationships, and help predict what will happen next. As a matter of fact, payers are already seeing huge benefits from incorporating AI into their business models, especially for combatting fraud and allocating resources.

It's important to step back and reiterate some of what we discussed at the start of this chapter. The United States spends a huge amount on healthcare, more than any other country both in total and per capita (Figure 10.2). However, all of that money doesn't buy better outcomes than other developed nations (Figure 10.3). So where does the money go? You won't be surprised to know that there's been a lot of research into exactly that.

In a seminal paper in *JAMA* in 2018, Papanicolas et al. examined the differences between the United States and other advanced economies in terms of their overall healthcare expenditures and the associated outcomes.[18] They found that back in 2016, the United States spent almost twice as much on medical care as ten high-income countries, despite a poorer performance when it came to population health. Social spending and healthcare usage didn't differ substantially between the United States and other high-income nations. The main drivers of the spending differences appeared to be the prices of labor and goods (including pharmaceuticals and devices) and administrative costs (Figure 10.4).

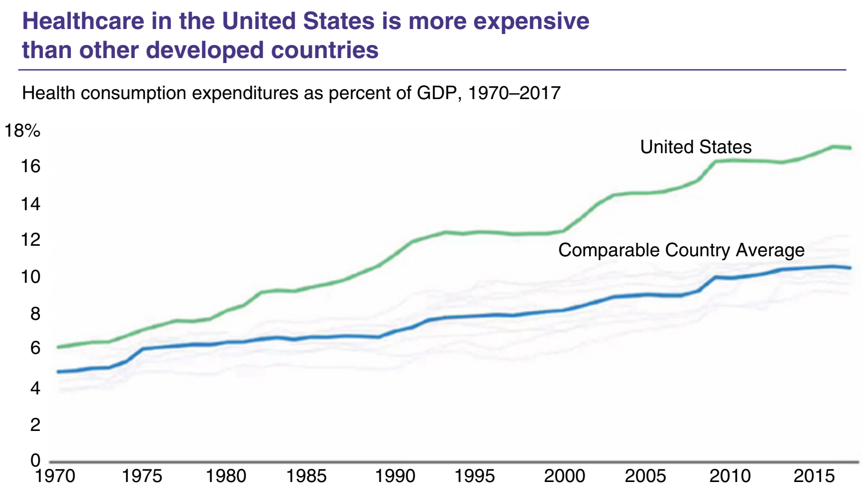

FIGURE 10.2 *(source: KFF Analysis of OECO and National Health Expenditure)*

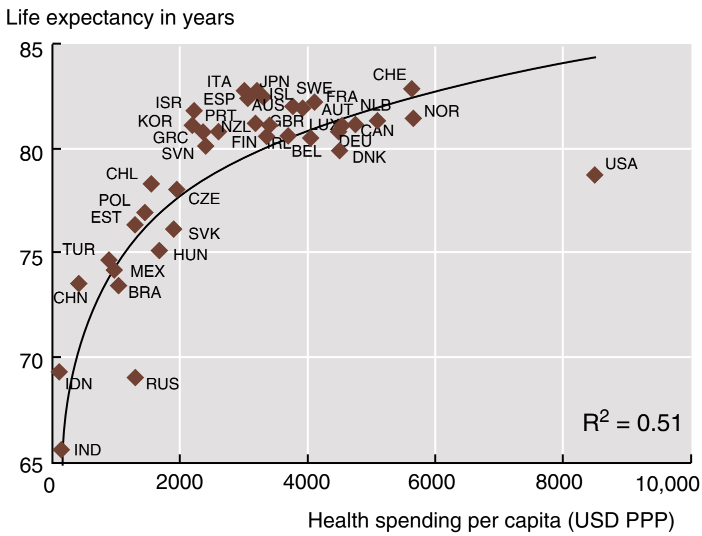

FIGURE 10.3 *(source: OECD Health Statistics 2013)*

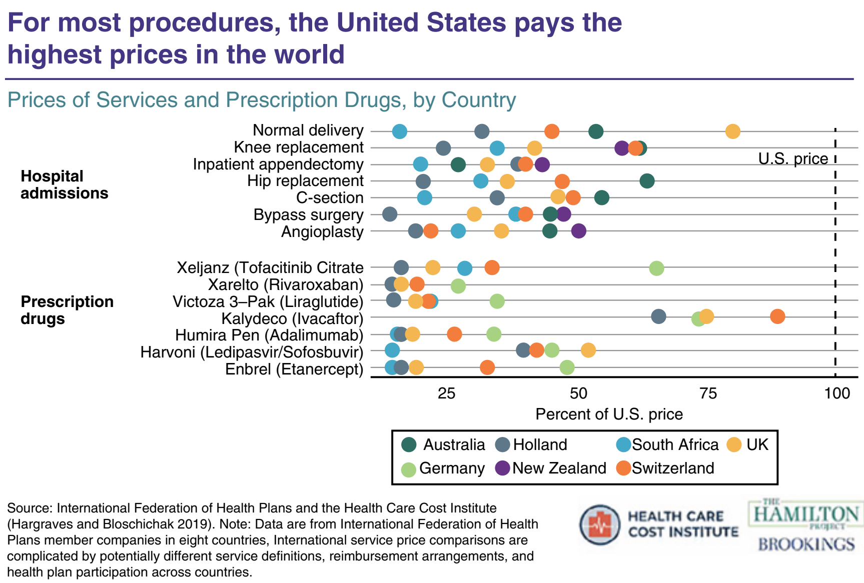

FIGURE 10.4 *(source: Health Care Cost Institute)*

10.2.1 Payer Administrative Functions

The administrative costs of healthcare (which are due to planning, regulating, and managing health systems) is higher in the United States than other advanced nations by a magnitude of 3–8, and a big reason for the United

States having much higher overall costs of care. We've already talked about some of the reasons for this, but the main one is the large number of insurance companies with their own benefit plan requirements (more than 1000 of them!), forms, and filing requirements.

It doesn't look like the number of insurance companies in the United States will shrink any time soon, and our public–private healthcare system is unlikely to become a single-payer system. So how can we lower administrative costs?

One option is to automate as much of it as we can. We've already talked about NLP, computer vision, and RPA, as well as how they're starting to make inroads into taking over administrative functions. The same is true for payers.

Attacking this problem from both sides makes the most sense. Jonathan Bush, the former CEO of AthenaHealth, wrote in 2018 that the greatest near-term opportunity for AI in healthcare isn't for headline-grabbing moonshots but for putting computers and algorithms to work on the most mundane drudgery possible.[19] Using AI to take care of back office and mundane activities that are also resource hungry and that lead to fatigue and burnout instead of better patient care would add a huge amount of value to the healthcare system.

Faxes are still a common method of communication between doctors' offices, payers, and providers, and processing these faxes uses a lot of manpower. Faxes don't use structured text, which is why it takes an average of two and a half minutes to review each document and translate it into inputs for EHRs. Machine learning and business process outsourcing have helped Athena Health to automate faxes and reduced processing time to 1 min and 11 s. This resulted in the elimination of over three million hours of work for one healthcare system. Athena has used AI to further reduce import time to just 30 s, and they've created software that can scan lab results and flag anything urgent that needs human attention. They're also working on an algorithm that could schedule routine follow-ups for high-risk patients. All of this can reduce administrative work for both payers and providers.

The idea of using EHRs to make doctors' jobs easier is yet to become a reality because of the need for manual data entry and how the design of EHRs makes information lookup and retrieval difficult. The next generation of technologies will help doctors. By making it easier to document using voice inputs, increasing coding accuracy, and filling out forms, the potential of intelligent automation could finally reach fruition. This will lessen the workload on the sides of both payers and providers because it's the back and forth between these two sides that generates so much administrative work.

According to a study by Accenture, AI could significantly improve operations and boost the bottom line.[20] One way to accomplish this is by automating processes across billing, customer service, claims, enrollment, and quality and compliance. The report concludes that automating functions could generate \$1.5 million in operating income for every hundred employees at payers. According to a different Accenture survey, 72% of executives at payers believe that AI will be among the top three strategic priorities for their company within the year. In the near term, they hope that AI will be used to anticipate and resolve questions from customers, accelerate the prior authorization and clinical review of claims, and improve the process of benefits loading and design.

The biggest potential savings could come from using AI to anticipate and respond to customer demands. AI could unlock $2.1 billion for health insurers in this one area alone, as well as a further $1.4 billion from managing membership and billing, $1.1 billion from managing and supporting reimbursement, $1 billion from managing networks and providers, $900 million from performing health management and $500 million from managing quality improvement and compliance.[20]

Managing customer interactions is the biggest slice of the pie, and AI could allow us to predict what customers will ask based on earlier calls and other data. It could then help a cell center rep to satisfy the customer's needs through a more targeted interaction.

Virtual agents could take care of the simpler interactions and answer questions without human intervention. We could use AI to anticipate needs and to provide just-in-time information based on whatever the customer is asking for. Accenture's analysis also showcases two other areas that insurers should focus on in the short term, which are accelerating prior authorization and the clinical review of claims and improving the process of benefits loading and design. If this sounds like what we discussed for providers, that's because it is. The administrative work in healthcare occurs between payers and providers, so technologies like NLP, computer vision, and AI-enabled RPA will help both sides with coding reviews, prior authorization, and form completion.

Accenture also provided proposals for AI and machine learning that could demonstrate an impact on operating income in the near term. For example, RPA could automatically approve prior authorization requests. In fact, Express Scripts is already doing this, helped along by their acquisition of eviCore for $3.6 billion in 2017.[20,21] AI could help employees focus on decision-making and those more creative processes that need human intervention and oversight. This could unlock capacity within the institution because payers won't need to scale future jobs if their roles can be automated by AI. As is the case in every part of the healthcare value chain, the key barriers include poorly defined governance, mismatched or poor quality vendors, and unsuitable existing infrastructure.

10.2.2 Fraud

AI could have a significant impact when detecting fraud in health plans. We've already discussed the idea of upcoding and how it's a common method of fraud in healthcare. There are well-documented cases of unusual coding patterns that emerge in health systems or in regions where certain diagnostics or treatments are being prescribed at a rate that doesn't fit the national pattern. In many cases, this turns out to be due to a scheme of kickbacks or upcoding by certain providers. Historically, health plans have detected these abnormal patterns months or years after they've taken place. This usually sets off investigations and attempts to recover those funds, costing health plans significant financial and human resources.

More recently, health plans have started using AI to monitor for fraud and detect abnormal activities before they release payments to providers. As such, they can avoid paying for fraudulent activities, along with the associated costs of investigation and recovery. Insurance company Highmark used AI to save over $260 million by cutting down on fraud, waste, and abuse in 2019 alone. They were able to save more than $850 million between 2015 and 2020.[22, 23]

Insurance organizations are already using AI to detect fraudulent activity. Highmark is tapping into the tech to limit the financial exposure that their customers are at risk of, since payments for fraudulent activities ultimately involve out-of-pocket costs for patients. The AI software used in fraud detection can quickly adapt to changing behavior and pick up on abnormalities much faster than traditional tools, which usually tap into a set of established rules to spot suspicious behavior. The superior performance of Highmark's fraud detection systems was confirmed by Change Healthcare when they reviewed Highmark's Payment Integrity programs. This review found that Highmark's program outperformed the industry standard, saving 10% of medical claims for group customers and nearly 33% more than other national payers.[23]

This is also beneficial to other parts of a health plan's business as they can communicate with their customers about their ability to detect and prevent fraud while simultaneously saving them money through lower premiums and better service.

In an Optum survey of health plan leaders, 43% said they believe that AI will help detect fraud, waste, and abuse in reimbursement.[24] CMS has also adopted AI and other advanced technologies in an effort to reduce Medicare fraud, waste, and abuse.[25] Figure 10.5 shows an example of an algorithm for fraud detection from a McKinsey report about using AI in healthcare.[26]

10.2.3 Personalized Communications

AI can help health plan providers to improve the frequency and personalization of communications between providers and their members. An overabundance of communication touchpoints leads to fatigue on the part of members, and they end up ignoring them. Some members have received more than 200 messages from their health plan in just the space of a year. Even with the best intentions, this excessive communication has led to a problem called "member abrasion," which damages engagement rates and can even cause people to distrust health plans.[27]

AI can help us to develop better health engagement strategies that can reestablish trust. The retail industry has shown that AI can be used to expand traditional data sets for personalized communication, and that this can lead to higher engagement rates. Several health engagement technologies have emerged that can capture data about contexts, channel preferences, behaviors, and values. This data can help us to further personalize communications

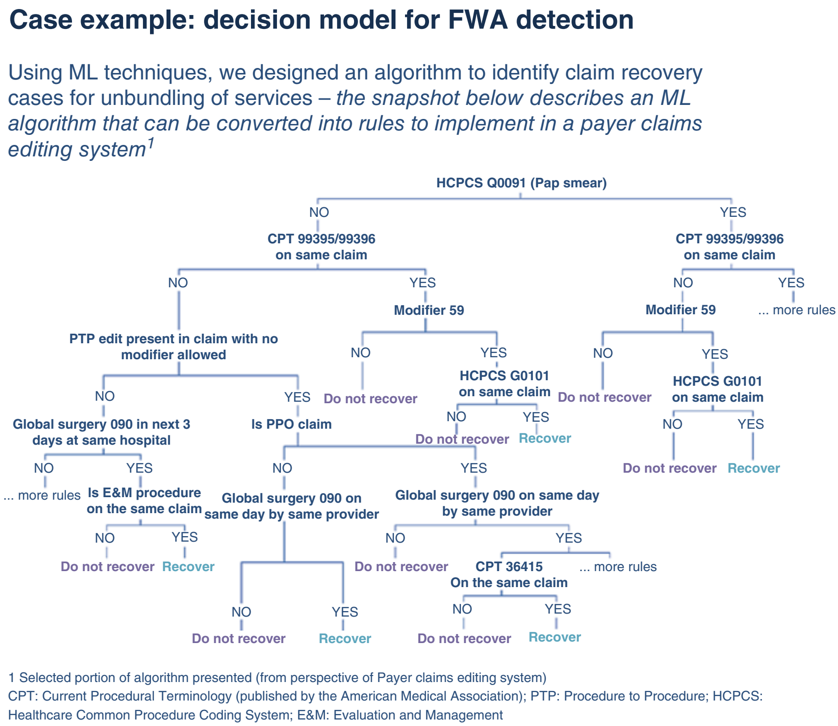

FIGURE 10.5 *(source: McKinsey & Company)*[26]

and lead to better health outcomes over time. These systems use AI and machine learning to track responses to campaigns, replicate the ones that are doing well and removing the ones that are failing. They can often do this in real time.

When done well, AI can humanize engagement, empowering healthcare consumers to participate in their own healthcare based on their responses to personalized touchpoints that are sent out in appropriate amounts and via the best channels for the consumer.

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